

Assessment 4.1: Home Safety and Conservation Quiz

Unit: Unit 4, Spaces We Live In (Environmental Design and Management) **Topic:** 4.1 Safe, Healthy, Sustainable Homes (Lessons 4.1 to 4.5) **When:** The last 15 minutes of Lesson 4.5 (day 97), starting at minute 23 of the period. Step 4 of that lesson's procedure holds the slot. **Category:** Topic assessments (25 percent). One retake after a review; best score counts. The retake review runs during Lesson 4.7 Day 1's team sort for students who need it. **Points:** 30 **Allowed:** a calculator. Nothing else. The unit word bank on the wall stays up. **Time:** 15 minutes. Extended time: finish in the first ten minutes of Lesson 4.6 or in a support period.

Home Safety and Conservation Quiz

Name: _____ Date: _____ Period: _____

Directions: Twelve items. Items 1 to 4 are matching; write the letter. Items 5 to 10 are short answer; a phrase is enough unless it says “sentence.” Item 11 is a scenario; write two or three sentences. Item 12 is a sort. Show your math on items 8 and 9. You may use a calculator.

Part A: Matching (items 1 to 4)

Match each hazard to the room where it is most common. Write the letter. Each room is used once.

Item	Hazard		Room
1. _____	Gasoline stored next to the water heater	A	Bathroom
2. _____	A pot handle turned out over the edge of the stove	B	Stairs
3. _____	A hair dryer plugged in with the cord in the sink	C	Garage
4. _____	A loose rug at the top with a curled corner	D	Kitchen

Part B: Short answer (items 5 to 10)

5. Where do smoke detectors go in a home, and how often do you test them? (two parts)



6. Why does a home need a carbon monoxide (CO) detector even if it has a smoke detector? One sentence.

7. A breaker trips. Write the two moves to reset it, in order. Then: it trips again right away. What do you do?

Move 1: _____ Move 2: _____ If it trips again: _____

8. Energy math. A 100 W bulb runs 4 hours a day for 365 days. Electricity costs \$0.25 per kWh.

$100 \times 4 \times 365 =$ _____ Wh. Divide by 1,000 = _____ kWh. Times \$0.25 = \$ _____ a year.

9. Water math. A faucet drips once a second, about 5 gallons a day. How many gallons in a year? A bathtub holds about 60 gallons. About how many bathtubs is that?

$5 \times 365 =$ _____ gallons. That is about _____ bathtubs.

10. A bathroom sink drains slowly. Write the first two things you do before you ever reach for a chemical drain cleaner.

First: _____ Second: _____

Part C: Scenario (item 11)

11. The Ferreira-Lund family lives in a two-story house: two parents, a 3-year-old, and a grandfather who uses a walker and sleeps on the second floor. Write their family emergency plan in two or three sentences. Your plan must include: two ways out of the grandfather's bedroom, one meeting spot that he can reach, who carries the 3-year-old, and the two numbers the family should know.





Part D: Sort (item 12)

12. Put each item in the bin that does the least harm. Write the letter of the bin next to each item.

Bins: R1 = Reduce (do not bring it in), R2 = Reuse, R3 = Recycle, C = Compost

Item	Bin
A banana peel	
A glass jar	
A stack of junk mail	
A plastic water bottle you have not bought yet	
A clean cardboard box you no longer need for storage	
A T-shirt with a hole	

Grade 8 stretch (no points): A low-flow showerhead costs \$15 and saves about \$70 a year. How many months until it pays for itself? _____

Teacher key

Total 30 points.

Item	Answer	Points	Standard line checked
1	C (Garage)	1	EDM 1. b); HSW 3. c)
2	D (Kitchen)	1	EDM 1. b); HSW 3. c)
3	A (Bathroom)	1	EDM 1. b); HSW 3. c)
4	B (Stairs)	1	EDM 1. b); HSW 3. c)
5	On every level of the home and in or near every sleeping area (1 point); test monthly (1 point). Accept “every floor and every bedroom.”	2	EDM 1. b); Std 2 PI B
6	Carbon monoxide has no smell (and no color), so nobody can notice it without a detector; a smoke detector does not sense CO. Accept any sentence with “no smell” or “cannot smell it.”	2	EDM 1. b); HSW 3. c)
7	Move 1: push it fully OFF (1). Move 2: push it ON (1). If it trips again: stop, unplug what you turned on, tell an adult; do not keep resetting it (1).	3	EDM 1. b); HSW 4. c); PSI 3. a) c., d., e.
8	146,000 Wh (1); 146 kWh (1); \$36.50 (1). Accept rounding to \$36 or \$37. If the divide step is missed but the method is otherwise	3	EDM 1. d); SUS 3. e); National FCS 3.4.3

	right, 1 point total.		
9	1,825 gallons (2); about 30 bathtubs (1). Accept 30 or 31.	3	EDM 1. d); SUS 3. e); National FCS 3.4.3
10	First: pull out the hair or gunk with a plastic drain tool (or by hand with gloves) (1). Second: plunge with water over the cup, then hot water (1). Order can be swapped if both are named. “Never mix chemicals” as a bonus note, no extra points.	2	EDM 1. b); PSI 3. a) d.; HSW 4. c)
11	Two ways out of the grandfather’s room: the door to the hall and the window (or a plan to wait at the window with the door closed) (1). A meeting spot he can reach with a walker: the mailbox, a neighbor’s front step, a fixed object close by; not the driveway (1). A parent carries the 3- year-old (1). 911 and Poison Control 1-800-222-1222 (1). Written as a plan, not a list (1).	5	EDM 1. c); HSW 3. e); HSW 4. g); Std 2 PI B
12	Banana peel C; glass jar R2; junk mail R3 (accept R1 with “sign off the list”); water bottle not yet bought R1; clean	6	EDM 1. d); SUS 2. c); National FCS 2.2.3

	cardboard box R3 (accept R2); T-shirt with a hole R2. One point each.		
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Stretch: $\$15 / \$70 = 0.21$ years, about 2.6 months; accept “about 3 months.” No points.

Scoring bands: 27 to 30 got it; 21 to 26 approaching, no retake needed unless the student wants one; 20 or below, retake after the review. The review for retakes runs during Lesson 4.7 Day 1’s team sort and covers items 7, 8, and 11, which are the ones most often missed.

Reading the results: if more than a third of the class misses item 8 on the divide step, run one more bulb problem as the Lesson 4.6 do now before the design topic starts. If item 11 is weak, the Design a Room client brief in Lesson 4.11 should ask every client’s plan for the same three parts.